

Heart Disease Prediction

Final Project Report — AI/ML Internship

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Project Title: Heart Disease Prediction

1. Problem Statement

Cardiovascular disease is among the leading causes of death worldwide, and early detection of at-risk patients can significantly improve treatment outcomes. The goal of this project is to build a machine learning model that predicts whether a patient is likely to have heart disease based on clinical measurements such as age, blood pressure, cholesterol level, and ECG results, and to make that model accessible through a simple, interactive web application.

2. Introduction

This project uses the widely-studied UCI Cleveland Heart Disease dataset, which contains 303 patient records described by 13 clinical features and a binary target indicating the presence or absence of heart disease. The project follows a standard supervised-learning workflow:

- Exploratory Data Analysis (EDA) to understand feature distributions and correlations.
- Data preprocessing: train/test split and feature scaling.
- Training and comparing five classification algorithms.
- Hyperparameter tuning of the best-performing model with GridSearchCV.
- Deploying the final model in an interactive Streamlit web application.

2.1 Dataset Description

Feature	Description
age	Age of the patient (years)
sex	Sex (1 = male, 0 = female)
cp	Chest pain type (0-3)
trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar > 120 mg/dl (1 = true)
restecg	Resting electrocardiographic results (0-2)
thalach	Maximum heart rate achieved
exang	Exercise-induced angina (1 = yes)
oldpeak	ST depression induced by exercise relative to rest
slope	Slope of the peak exercise ST segment (0-2)
ca	Number of major vessels colored by fluoroscopy (0-4)
thal	Thalassemia (1 = normal, 2 = fixed defect, 3 = reversible defect)
target	1 = heart disease present, 0 = no heart disease

3. Exploratory Data Analysis

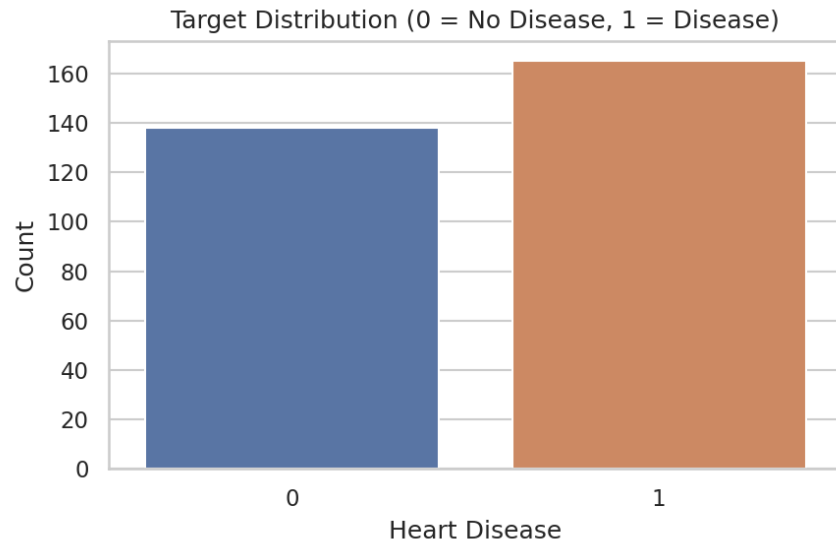


Figure 1: Distribution of the target class — moderately balanced (165 vs 138).

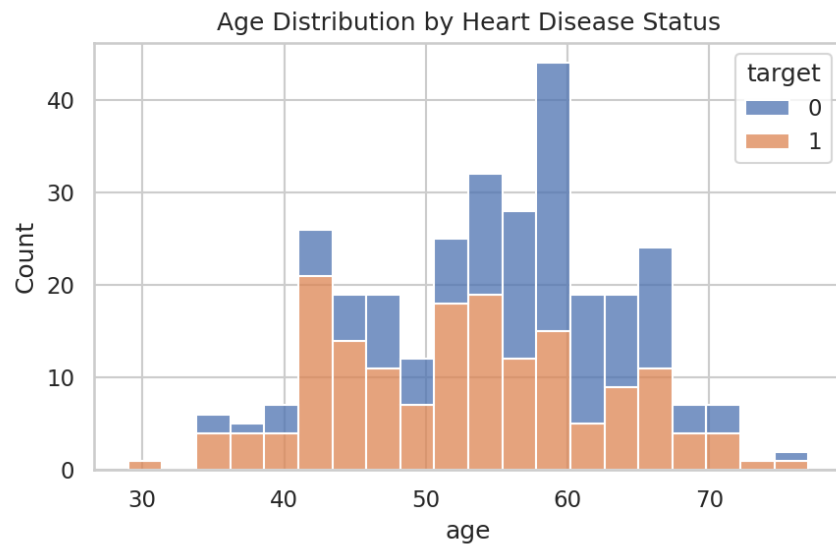


Figure 2: Age distribution split by heart disease status.

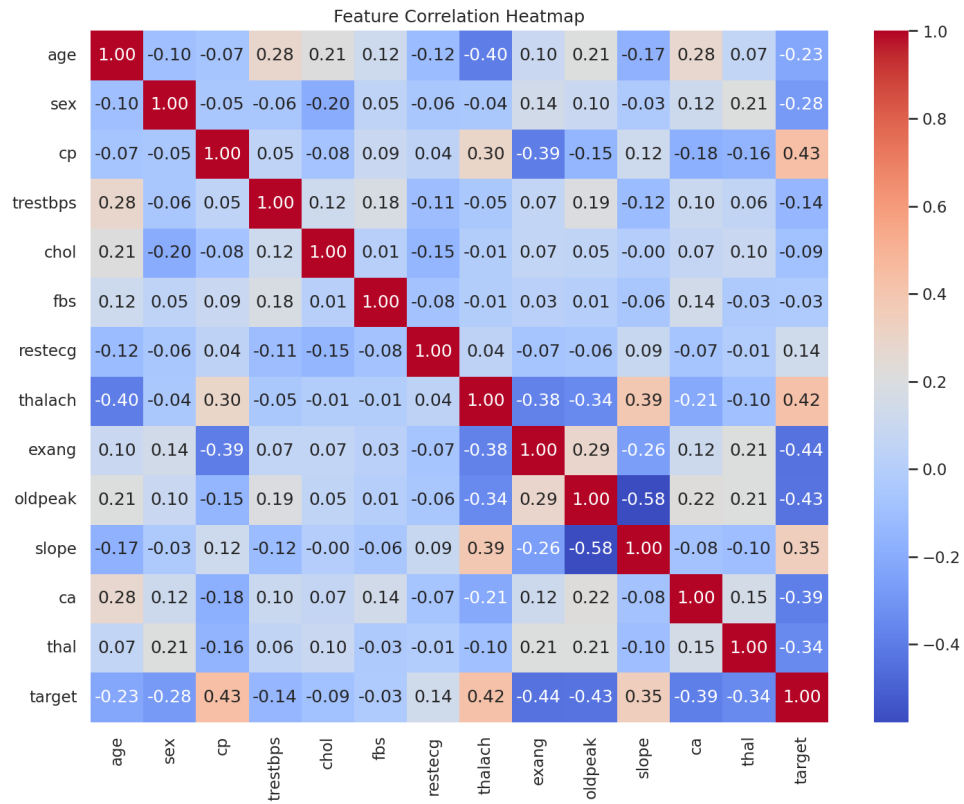


Figure 3: Correlation heatmap of all features.

4. Methodology

The data was split into training (80%) and test (20%) sets using stratified sampling to preserve class balance. All features were standardized using **StandardScaler**. Five classifiers were trained and evaluated using 5-fold cross-validation: Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Support Vector Machine (RBF kernel). The best-performing model was then tuned further using **GridSearchCV**.

5. Results

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	CV Mean Accuracy	CV Std
Random Forest	0.82	0.762	0.97	0.853	0.912	0.835	0.043
SVM (RBF)	0.82	0.775	0.939	0.849	0.883	0.802	0.039
Logistic Regression	0.803	0.769	0.909	0.833	0.869	0.831	0.041
K-Nearest Neighbors	0.803	0.769	0.909	0.833	0.863	0.802	0.027
Decision Tree	0.705	0.703	0.788	0.743	0.698	0.732	0.04

Table 1: Comparison of all trained models on the held-out test set.

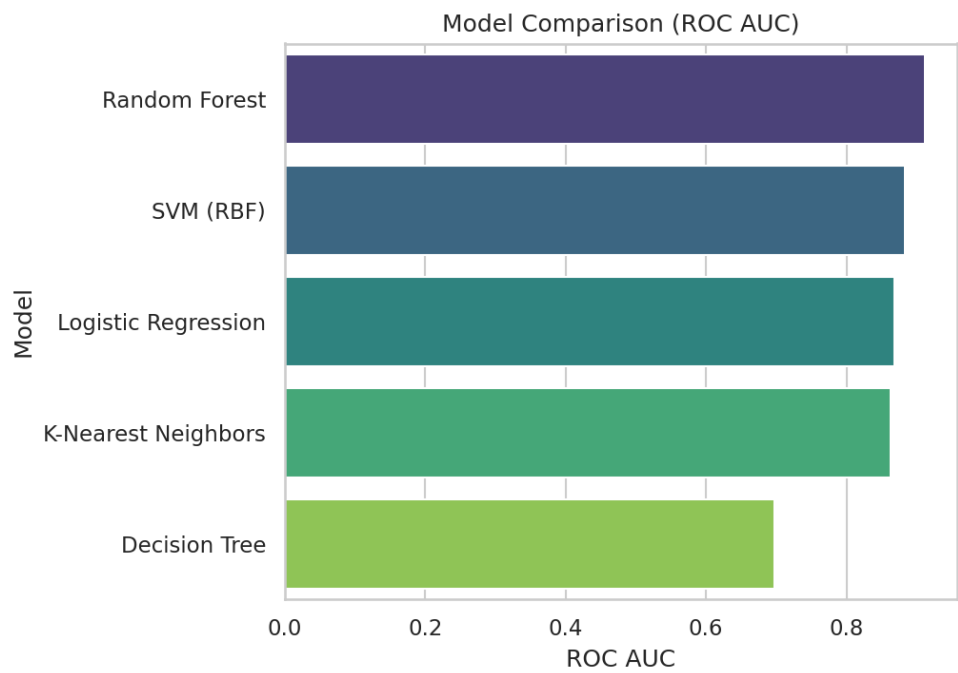


Figure 4: Model comparison by ROC AUC.

5.1 Final Tuned Model

The **Random Forest** classifier achieved the best ROC AUC and was selected as the final model. After hyperparameter tuning with GridSearchCV, the best parameters found were: **{'max_depth': 10, 'min_samples_split': 2, 'n_estimators': 200}**.

Metric	Value
Accuracy	0.820
Precision	0.762
Recall	0.970
F1 Score	0.853
ROC AUC	0.907

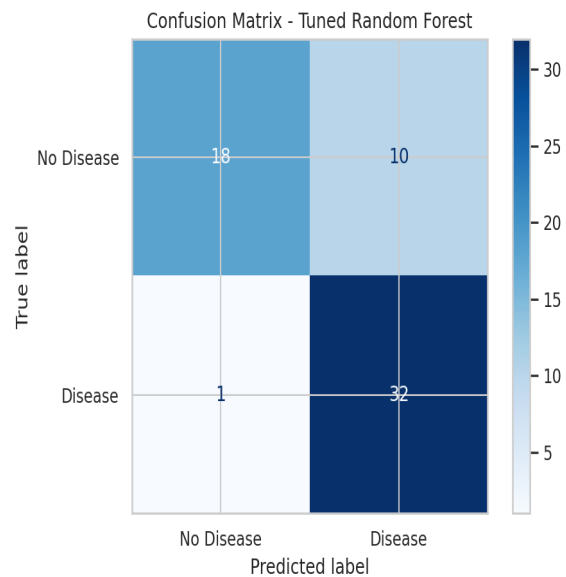


Figure 5: Confusion matrix of the tuned Random Forest on the test set.

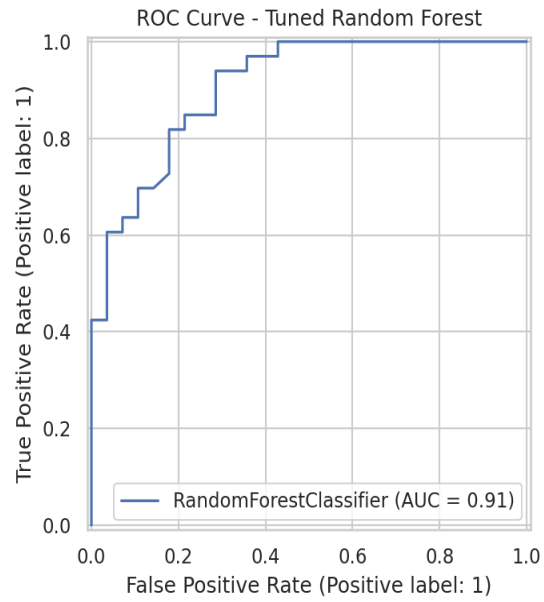


Figure 6: ROC curve of the tuned Random Forest.

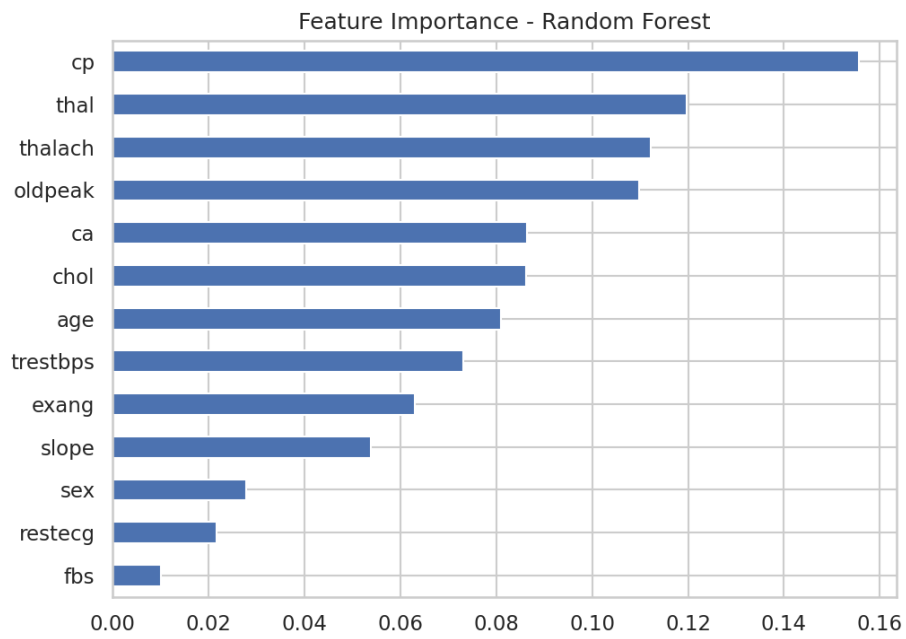


Figure 7: Feature importance ranking from the Random Forest model.

6. Application UI

The final model is served through a Streamlit web application that lets a user enter a patient's clinical details and receive an instant prediction with an estimated probability of heart disease.

Deploy

 **Heart Disease Prediction**

Enter the patient's clinical details below. This tool uses a machine learning model (trained on the UCI Cleveland Heart Disease dataset) to estimate the likelihood of heart disease. **This is an educational demo, not a medical diagnostic tool.**

Age	Max Heart Rate Achieved
54 - +	150 - +
Sex	Exercise Induced Angina
Male ▾	No ▾
Chest Pain Type	ST Depression (oldpeak)
Typical Angina ▾	1.00 - +
Resting Blood Pressure (mm Hg)	Slope of Peak Exercise ST Segment
130 - +	Upsloping ▾
Serum Cholesterol (mg/dl)	Number of Major Vessels (0-4)
246 - +	0 ▾
Fasting Blood Sugar > 120 mg/dl	Thalassemia
No ▾	Normal ▾
Resting ECG Result	
Normal ▾	
<button>Predict</button>	

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Figure 8: Application input form.

Heart Disease Prediction

Enter the patient's clinical details below. This tool uses a machine learning model (trained on the UCI Cleveland Heart Disease dataset) to estimate the likelihood of heart disease. **This is an educational demo, not a medical diagnostic tool.**

Age	Max Heart Rate Achieved
54	150
Sex	Exercise Induced Angina
Male	No
Chest Pain Type	ST Depression (oldpeak)
Typical Angina	1.00
Resting Blood Pressure (mm Hg)	Slope of Peak Exercise ST Segment
130	Upsloping
Serum Cholesterol (mg/dl)	Number of Major Vessels (0-4)
246	0
Fasting Blood Sugar > 120 mg/dl	Thalassemia
No	Normal
Resting ECG Result	
Normal	
Predict	

Result

⚠️ High likelihood of heart disease — estimated probability: 62.8%

This prediction is generated by a statistical model for educational purposes only and should never replace professional medical advice.

Figure 9: Application prediction result.

7. Future Approach

- Collect a larger and more demographically diverse dataset to improve generalization beyond the 303-record Cleveland cohort.
- Add model explainability (e.g. SHAP values) so each individual prediction can be interpreted feature-by-feature.
- Experiment with gradient boosting methods (XGBoost, LightGBM) to see if they outperform Random Forest.
- Wrap the model in a REST API (e.g. FastAPI) so it can be integrated into other clinical or research systems.
- Add authentication and prediction history/logging for longitudinal patient tracking.
- Deploy with continuous monitoring for model drift as new patient data becomes available.

8. Conclusion

The final tuned Random Forest model achieved an accuracy of 82.0% and an ROC AUC of 0.907 on the held-out test set, demonstrating that clinical measurements can meaningfully predict the presence of heart disease. The accompanying Streamlit application makes this model accessible through a simple, user-friendly interface, fulfilling the goals of this internship project.

Disclaimer: This project is for educational purposes only and is not intended for real medical diagnosis.